

Exp No:5 Data Collection and Initial Exploration

```
import pandas as pd
db = pd.read_csv('diabetes.csv')
print(db.head())
print(db.info())
print(db.describe())
import matplotlib.pyplot as plt
import seaborn as sns
db.hist(bins=50, figsize=(20, 15))
plt.show()
sns.pairplot(db)
plt.show()
```

Output:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72.0	35	169.5	33.6
1	1	85	66.0	29	102.5	26.6
2	8	183	64.0	32	169.5	23.3
3	1	89	66.0	23	94.0	28.1
4	0	137	40.0	35	168.0	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

<class 'pandas.DataFrame'>

RangeIndex: 768 entries, 0 to 767

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	float64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	float64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

dtypes: float64(4), int64(5)

memory usage: 54.1 KB

None

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	121.677083	72.389323	29.089844	141.753906
std	3.369578	30.464161	12.106039	8.890820	89.100847
min	0.000000	44.000000	24.000000	7.000000	14.000000
25%	1.000000	99.750000	64.000000	25.000000	102.500000
50%	3.000000	117.000000	72.000000	28.000000	102.500000
75%	6.000000	140.250000	80.000000	32.000000	169.500000
max	17.000000	199.000000	122.000000	99.000000	846.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	32.434635	0.471876	33.240885	0.348958
std	6.880498	0.331329	11.760232	0.476951

min	18.200000	0.078000	21.000000	0.000000
25%	27.500000	0.243750	24.000000	0.000000
50%	32.050000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000



